Chemistry - Undergraduate Level Assessment 16 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. What is the NMR frequency of 31 P in a 20.0 T magnetic field?
   - (a) 54.91 MHz
   - (b) 239.2 MHz
   - (c) 345.0 MHz
   - (d) 2167 MHz
2. Which of the following statements about the lanthanide elements is NOT true?
   - (a) The most common oxidation state for the lanthanide elements is +3.
   - (b) Lanthanide complexes often have high coordination numbers (> 6).
   - (c) All of the lanthanide elements react with aqueous acid to liberate hydrogen.
   - (d) The atomic radii of the lanthanide elements increase across the period from La to Lu.
3. Which of the following statements most accurately explains why the 1 H spectrum of 12 CHCl 3 is a singlet?
   - (a) Both 35 Cl and 37 Cl have I = 0.
   - (b) The hydrogen atom undergoes rapid intermolecular exchange.
   - (c) The molecule is not rigid.
   - (d) Both 35 Cl and 37 Cl have electric quadrupole moments.
4. Which of the following is always true of a spontaneous process?
   - (a) The process is exothermic.
   - (b) The process does not involve any work.
   - (c) The entropy of the system increases.
   - (d) The total entropy of the system plus surroundings increases.
5. Calculate the equilibrium polarization of 13 C nuclei in a 20.0 T magnetic field at 300 K.
   - (a) 10.8 x $10^{-}5$
   - (a) 4.11 x $10^{-}5$
   - (a) 3.43 x $10^{-}5$
   - (a) 1.71 x $10^{-}5$

True / False (10 marks)

Answer True or False

6. True or False: The thermal stability of group-14 hydrides increases in the order: CH 4 < SiH 4 < GeH 4 < SnH 4 < PbH 4.

7. True or False: The orbital angular momentum quantum number, l, of the electron most easily removed from ground-state aluminum is 2.

8. True or False: The +1 oxidation state is more stable than the +3 oxidation state for thallium (Tl).

9. True or False: The magnetogyric ratio of 23 Na is 7.081 x 10^7 $T^{-}1s^{-}1$.

9. True or False: The difference in infrared absorption frequency between DCl and HCl is primarily due to their reduced mass.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of metal ions as paramagnetic quenchers in chemical reactions, specifically analyzing why Cr_3+ is not suitable for this purpose.

12. Discuss the solid-state structures of the principal allotropes of elemental boron, focusing on the role of B_{12} icosahedra in their formation and stability.

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